

Presented by Idea Khan

ADAPTIVE RETRIEVAL GATING FOR MEDICAL QA (VARIANT 5)

MOTIVATION: ADDRESSING THE 'TRANSLATION GAP' IN RADIOLOGY

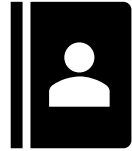
KEY ISSUES	ISSUES	DETAILS
Understanding	Jargon-Heavy reports	Complex terminologies and medical jargons -> leads to anxiety + struggle amount pateints to comprehend their own health information
Safety-exposure	PHI Exposure risk in Cloud RAG	RAG raises concerns regarding exposure of PHI (Protected Health Information)
Offline privacy-gates	Demand for Privacy-gated appliances	Offline solutions which prioritises patient privacy -> no exposure to unauthorized entities.

GOALS & RESEARCH QUESTIONS

- **Benchmark Privacy-gated retrievals:** Establish performance metrics, comparing privacy-gated RAG against traditional baselines. Measure effectiveness of innovative approaches
- **Quantify Accuracy-per-Watt:** Aim to measure & optimize balance between accuracy & energy efficiency, deliver insights into env. impact of medical architecture
- **Deployment guidelines for NHS and LMIC:** Craft guidelines & frameworks to implement privacy-preserving tech. tailored for healthcare settings in NHS and Low-middle Income countries



SYSTEM PIPELINE OVERVIEW



Raw Radiology report input

Initiate system with unprocessed inputs -> foundation for transformation into user-friendly formats



PHI detecting & Gating

Employ advance algo. To detect and obfuscate PHI, ensuring compliance with privacy regulations



Selective Local Retrieval

Implement method to extract information, optimize contextual retrieval databases depending on input



Contextualized generation of report

RAG model alongside Agents to generate accessible summary on findings with reliable citations for patients reports

MAIN DATABASE: CORE CORPORA AND THE MEDCORP DATABASE

MIMIC-CXR Radiology Database:

A publicly available dataset of 227,835 chest radiographs and free-text radiology reports.

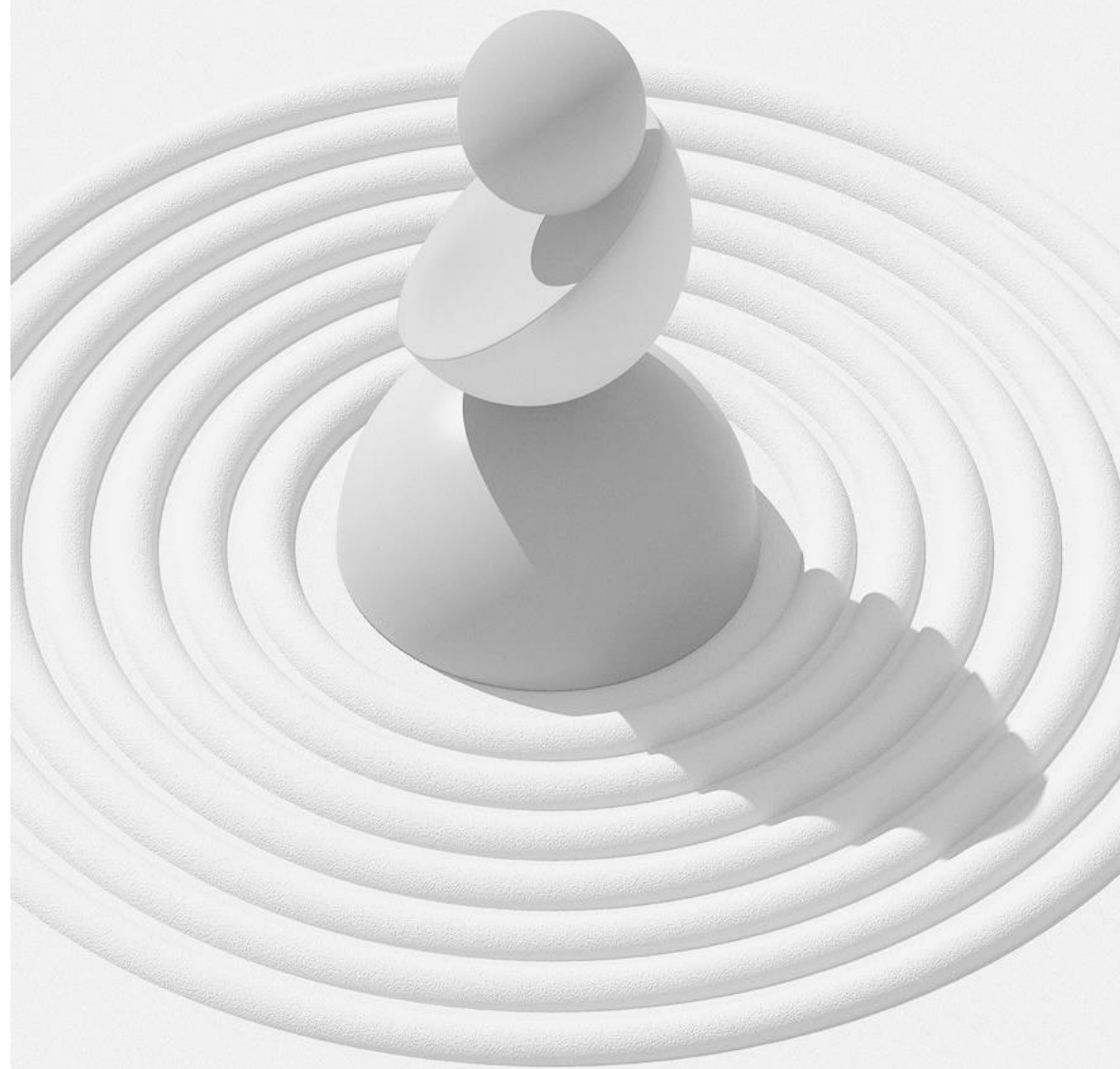
Use Case: Ideal for testing the "Contextualizer" module

MedCorp (The MedRAG Corpus):

A large-scale, comprehensive collection designed specifically for medical Question Answering

Usecase: Benchmarks show that RAG systems using MedCorp consistently outperform standard CoT

<https://huggingface.co/MedRAG>



SELECTIVE RETRIEVAL AND LOCAL INDEXING MECHANISMS

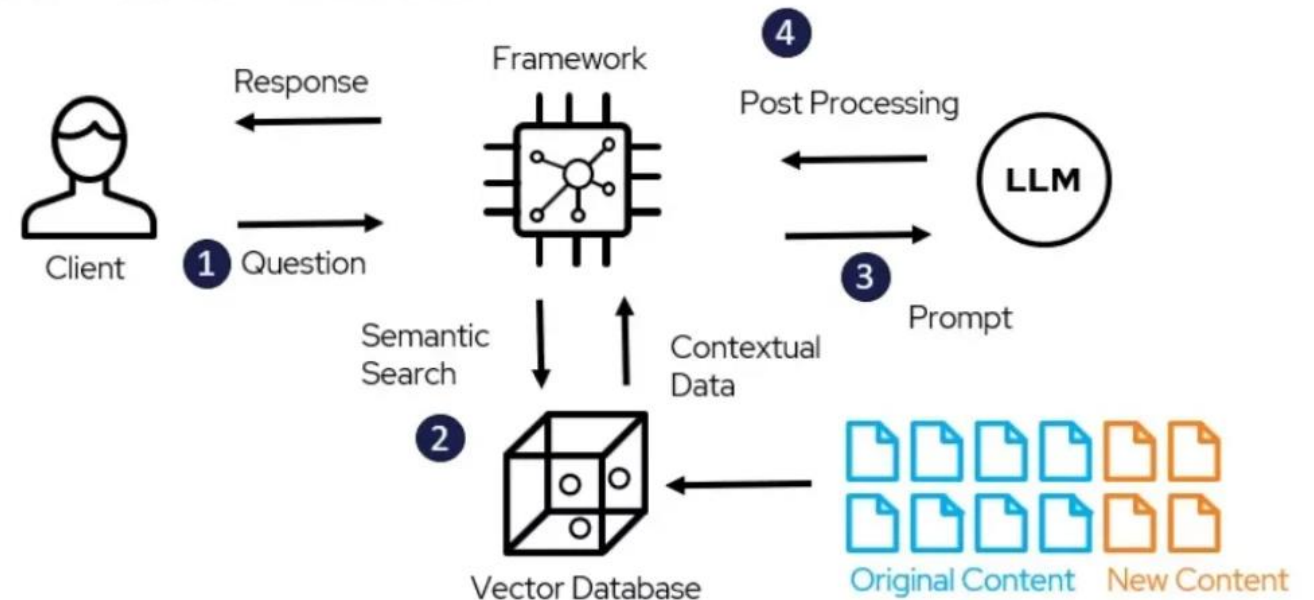
Hybrid BM25 and FAISS Dense Search

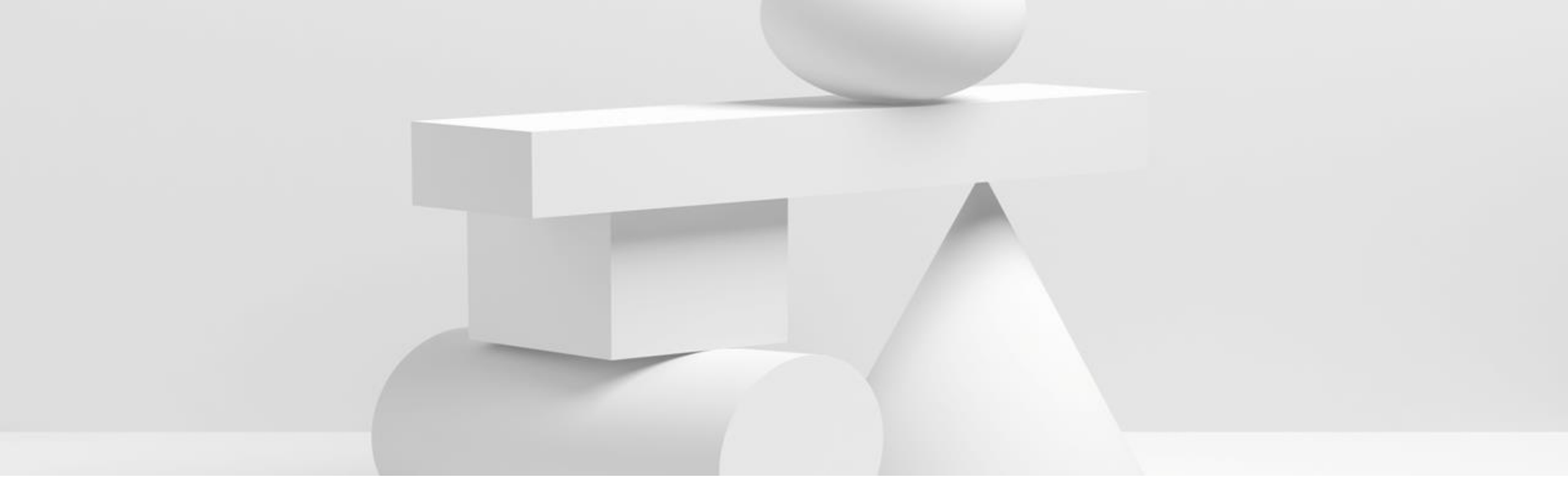
- Blends classic & FAISS to enhance accuracy and retrieval speed

Curated Corpus for enhanced contextualization

- Create a solid foundational material for the contextualization process consisting of research papers + clinical resources

RAG Architecture Model





RESEARCH + RETRIEVAL FRAMEWORKS

Privacy-Preserving RAG on Local Devices (Gemma 3):

- Establishes the feasibility of running a medical RAG framework entirely offline on consumer-grade hardware -> ensures GDPR compliance

BENCHMARKING HARNESS

1. Energy & latency profiling:

➔ zues-apple-silicon = Python library specialized for energy per-core consumption measuring the “Accuracy-per-watt” metric

2. Kiln AI

➔ Interactive RAG evaluation set using synthetic questions.
(MEDAES QA dataset)

RETRIEVAL POLICIES

1. Always-Retrieve | BM25 + vector every query
2. Always-Retrieve + Citation | P1 + source citation extraction
3. Closed-Book | None - parametric knowledge only
4. Hybrid Retrieval | BM25 vs. vector routing + RRF
5. **Selective/Gated** | Training-free gate decides per-query
6. User-Choice Toggle | User selects Fast vs. Sourced | HCI component |

P95 latency

95% of user requests are completed faster than a specific time, while the slowest 5% take longer

BENCHMARK RESOURCES

- <https://neurips.cc/virtual/2024/poster/95135>
- <https://medium.com/@chinmayd49/rag-production-optimizations-and-trade-offs-a623e5834e65>
- <https://arxiv.org/html/2601.02522v1>
- <https://pmc.ncbi.nlm.nih.gov/articles/PMC12739145/>

PROJECT TIMELINE: MONTH 1 & 2

Literature review on PHI Masking:

Review on current methodologies regarding PHI to develop a privacy-grading module

FAISS index construction:

Build FAISS index using data from PubMed & MedCorp -> rapid retrieval processes for system

Implement Gating Modules:

Develop & Integrate gating module including citation contract ensure data compliance and reliability



PROJECT TIMELINE: MONTH 3 & 4

Multi-device Energy & latency runs:

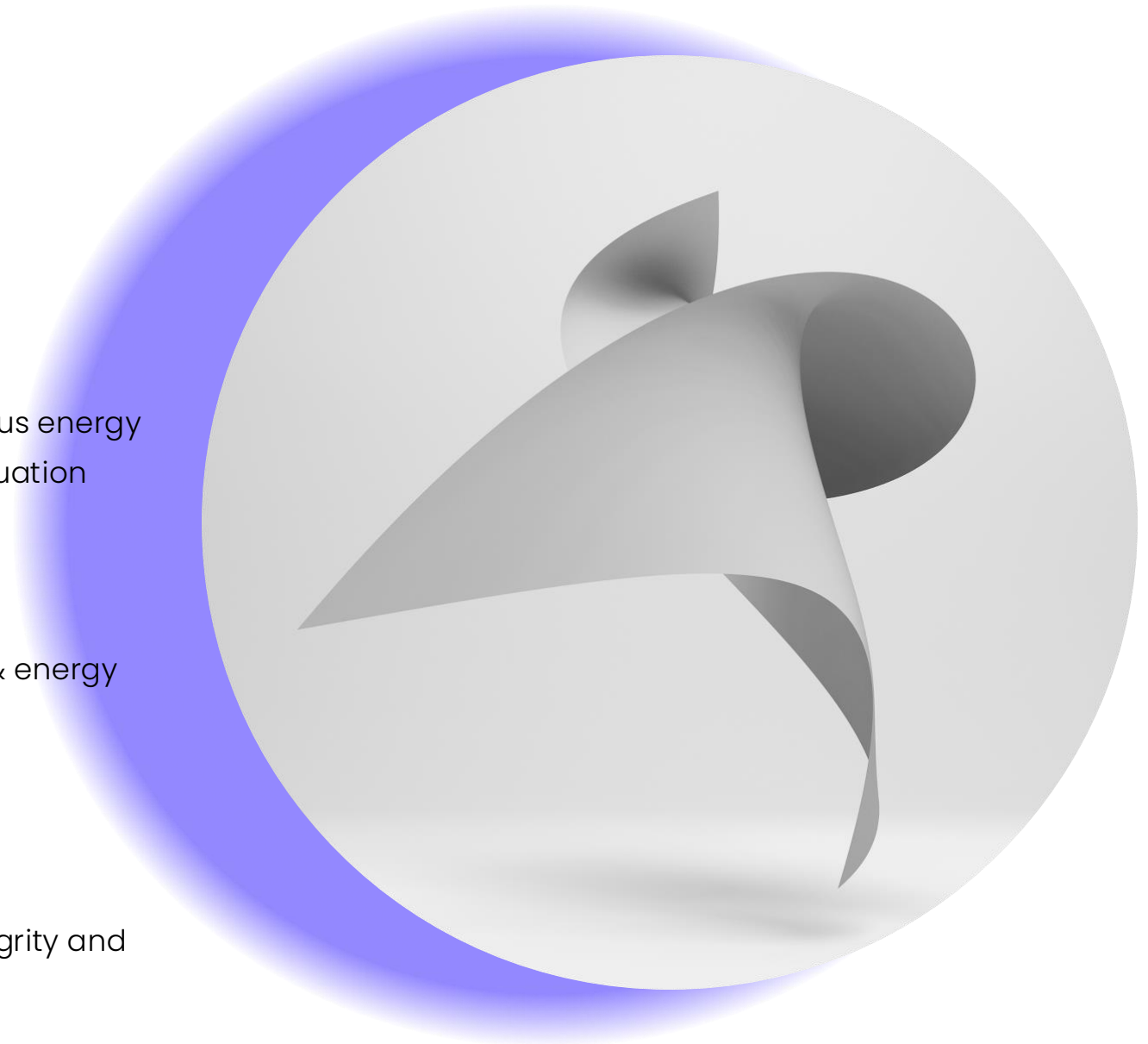
Conduct benchmarking through Virtual machines with various energy and hardware computations to ensure comprehensive evaluation

Trade-off curve analysis

Engage in detailed analysis of trade-off between accuracy & energy efficiency. Refine model for best operational cost.

Report generation and testing

Optimize and test patient report generation, verifying its integrity and possible application for real-world use.





THANK YOU!